INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Fourth Semester Examination, March - 2025

Subject Code: BCACC403

Subject Name: Operating Systems

Full Marks: 30 | Set No: 1 | Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. Define the critical section and explain its importance in process synchronization. **2**

2. What is a semaphore? How does it help in synchronization? **2**

3. Define Deadlock. **2**

4. What is memory fragmentation? **2**

5. Explain the concept of mutual exclusion with an example. **2**

6. List and explain the necessary conditions for deadlock. **2**

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. Discuss the role of monitors in process synchronization. **5**

8. Explain the Banker's algorithm for deadlock avoidance with an example. **5**

9. Describe paging and segmentation in memory management. **5**

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. Explain the different deadlock handling techniques. **10**

11. Describe the different memory management techniques used in operating systems. **10**

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Subject Code: BCACC403

Subject Name: Operating Systems

Full Marks: 30 | Set No: 2 | Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. Define Mutex. **2**

2. What is binary semaphore? **2**

3. Explain the concept of paging in memory management. **2**

4. Write an example of software-based techniques of mutual exclusion in synchronization. **2**

5. Describe Resource Allocation Graph. **2**

6. Define starvation and how it differs from deadlock. **2**

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. Analyze how demand paging improves memory utilization. **5**

8. Differentiate between internal and external fragmentation. **5**

9. What is page and frame? What the relationship is between page size and frame size? **5**

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. Create the Safe Sequence of the following Snapshot at time T_0: for 5 processes P_0 through P_4 having 3 resource types A (10 instances), B (5instances, and C (7 instances). **10**

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3 .	

11. Discuss in detail how virtual memory works and its advantages. **10**

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Subject Code: BCACC403

Subject Name: Operating Systems

Full Marks: 30 | Set No: 3 | Duration: 1 Hour

Part - A

Answer any 5 out of 6 questions

Each question carries 2 marks (2 × 5)

1. Define Livelock. 2

2. What is general semaphore? 2

3. What is MMU in memory management. 2

4. What is Safe State? 2

5. What is circular-wait? 2

6. What are the benefits of memory management? 2

Part - B

Answer any 2 out of 3 questions

Each question carries 5 marks (5 × 2)

7. Differentiate between deadlock prevention and deadlock avoidance. 5

8. What are the classical problems of Synchronization explain them. 5

9. Illustrate the structure of the Page Table. 5

Part - C

Answer any 1 out of 2 questions

Each question carries 10 marks (10 × 1)

10. Illustrate why Internal Fragmentation and External Fragmentation occurs with proper example. 10

11. Discuss the Producer-Consumer problem in detail. How can it be solved using semaphores? Explain with an example. 10